COMPANION NOTES TO *WHY* $T - V$ *?*

From Calculus to the Calculus of Variations

A patient route to the Euler–Lagrange equation

Functions, functionals, perturbations, and stationary action

The idea in one sentence. Ordinary calculus asks what happens when we change a *number* slightly. The calculus of variations asks what happens when we change an *entire function* slightly.

These notes are intended to be read slowly. We shall first become comfortable with the objects we are differentiating. Only then will we derive an equation of motion. The aim is understanding, not a collection of techniques.

1 A function takes a number to a number

A familiar function might be

$$f(x) = x^2.$$

We give it a number x; it returns another number $f(x)$. For example, $f(3) = 9$.

In ordinary calculus we investigate what happens if we change the input a little:

$$x \longrightarrow x + \delta x.$$

The output changes by

$$f(x + \delta x) - f(x) = (x + \delta x)^2 - x^2 = 2x\,\delta x + (\delta x)^2.$$

When δx is very small, the squared term is much smaller than the linear term. The leading change is therefore

$$\delta f \approx 2x\,\delta x.$$

The derivative $f'(x) = 2x$ tells us how sensitive the output is to a small change in its input.

2 A functional takes a function to a number

Now let the input be not a number, but a *whole function* q.

For example, let q be any bounded real-valued function on $[0, 1]$. We can assign it the number

$$\boxed{\|q\|_\infty = \sup_{0 \le t \le 1} |q(t)|.}$$

This is the **supremum norm**. It measures the largest magnitude of the function (or, if the largest value is not attained, its least upper bound).

For $q(t) = t^2$, the supremum norm is 1. For $q(t) = \sin(2\pi t)$, it is also 1. Very different functions can have the same norm.

The map

$$q \longmapsto \|q\|_\infty$$

is a **functional**: it takes an entire function and returns a single number.

A useful distinction. A function such as $f(x) = x^2$ accepts a number. A functional such as $J[q] = \|q\|_\infty$ accepts a function. The square brackets $J[q]$ remind us that the input is the whole curve.

3 Other functionals: the whole curve matters

A functional need not be a norm. Here are two simple examples, for functions on $[0, 1]$:

$$J_1[q] = \int_0^1 q(t)\,\mathrm{d}t, \qquad J_2[q] = \int_0^1 q(t)^2\,\mathrm{d}t.$$

The first records signed area under the graph. The second records accumulated squared magnitude.

For $q(t) = t$, we obtain

$$J_1[q] = \frac{1}{2}, \qquad J_2[q] = \frac{1}{3}.$$

The input is still the *entire function*. The integral merely turns that function into one number.

Later, the action will be another functional:

$$S[q] = \int_{t_0}^{t_1} L(q(t), \dot{q}(t), t)\,\mathrm{d}t.$$

We need not understand the symbols in this last formula yet. For now, notice its shape: *give me a possible trajectory, and I will assign it a number.*

4 What does it mean to perturb a function?

In ordinary calculus, we change x to $x + \delta x$.

If our input is a function $q(t)$, we must change the *function*. We do so by adding another small function:

$$\boxed{q(t) \longrightarrow q(t) + \delta q(t).}$$

At each time t, the quantity $\delta q(t)$ tells us how far the original graph is displaced vertically.

A particularly clean way to express this is

$$\boxed{q_\varepsilon(t) = q(t) + \varepsilon\eta(t).}$$

Here:

- $q(t)$ is the original curve;
- $\eta(t)$ is the *shape* of the perturbation;
- ε is a small number controlling its size.

Thus $\delta q(t) = \varepsilon\eta(t)$. Holding η fixed while varying ε lets us use familiar one-variable calculus.

For example, if $q(t) = t$ and $\eta(t) = t(1 - t)$, then

$$q_\varepsilon(t) = t + \varepsilon t(1 - t).$$

The curve bows slightly away from the straight line, but the endpoints remain unchanged because $\eta(0) = \eta(1) = 0$.

The central visual picture. Ordinary differentiation nudges a point on a number line. Variational differentiation nudges an entire graph, while we keep track of the shape and size of that nudge.

5 A side-by-side comparison

Ordinary calculus	**Calculus of variations**
Input: a number x	Input: a function $q(t)$
Output: $f(x)$	Output: $J[q]$
Perturb: $x \mapsto x + \varepsilon h$	Perturb: $q \mapsto q + \varepsilon\eta$
Look at $f(x + \varepsilon h)$	Look at $J[q + \varepsilon\eta]$
Differentiate with respect to ε at 0	Differentiate with respect to ε at 0
At a stationary point: $f'(x) = 0$	At a stationary function: first variation is zero for every allowed η

The essential trick is that, although J takes a function as input, the expression $J[q + \varepsilon\eta]$ is an ordinary function of the single number ε.

6 Differentiating a functional: our first example

Take the smooth functional

$$J[q] = \frac{1}{2}\int_0^1 q(t)^2\,\mathrm{d}t.$$

Perturb the function: $q \mapsto q + \varepsilon\eta$. Then

$$\begin{aligned} J[q+\varepsilon\eta] &= \frac{1}{2}\int_0^1 (q+\varepsilon\eta)^2\,\mathrm{d}t \\ &= \frac{1}{2}\int_0^1 q^2\,\mathrm{d}t + \varepsilon\int_0^1 q\eta\,\mathrm{d}t + \frac{\varepsilon^2}{2}\int_0^1 \eta^2\,\mathrm{d}t. \end{aligned}$$

Compare this with the elementary expansion

$$(x+\varepsilon h)^2 = x^2 + 2\varepsilon x h + \varepsilon^2 h^2.$$

The structure is the same: original value, a term linear in ε, and a term quadratic in ε.

The coefficient of the linear term tells us the first-order response:

$$\boxed{\delta J[q;\eta] = \frac{\mathrm{d}}{\mathrm{d}\varepsilon} J[q+\varepsilon\eta]\Big|_{\varepsilon=0} = \int_0^1 q(t)\eta(t)\,\mathrm{d}t.}$$

This is called the **first variation**. It depends on the original function q *and* on the direction η in which we perturb it.

Notice that we have not differentiated with respect to t. We differentiated with respect to the perturbation size ε.

6.1 Why not differentiate the supremum norm first?

The supremum norm was a good example of a functional, but it is not generally differentiable. Even the simple one-variable expression $|x|$ has a corner at $x = 0$. The supremum norm has analogous difficulties. That is why our first derivative example uses a smooth integral functional.

7 What is a stationary function?

In ordinary calculus, x_* is stationary for f if

$$f'(x_*) = 0.$$

In the calculus of variations, q_* is stationary for J if

$$\boxed{\delta J[q_*;\eta] = 0 \quad \text{for every allowed perturbation } \eta.}$$

Why *every* perturbation? Because a curve can be altered in infinitely many different ways: near the beginning, near the middle, over a broad interval, or in a narrow bump.

Being insensitive to just one particular bump would not be enough. Stationarity means no allowed small change has a nonzero *first-order* effect.

As in ordinary calculus, stationary does not necessarily mean minimum. It could mean maximum or another kind of stationary point.

8 From a functional to a trajectory

Now imagine that $q(t)$ describes a particle's position over time. Its velocity is $\dot{q}(t)$.

We want to compare different possible trajectories that share the same starting and finishing positions:

$$q(t_0) = q_0, \qquad q(t_1) = q_1.$$

If we perturb the path by $q_\varepsilon = q + \varepsilon\eta$, the endpoints must remain fixed. Therefore

$$\boxed{\eta(t_0) = \eta(t_1) = 0.}$$

Between the endpoints, $\eta(t)$ is free to take any sufficiently smooth shape.

The action functional is

$$\boxed{S[q] = \int_{t_0}^{t_1} L(q, \dot{q}, t)\,\mathrm{d}t.}$$

The quantity L is called the **Lagrangian**. It can depend on position, velocity, and time.

In the previous notes, we motivated the familiar mechanical choice

$$L = T + W = T - V,$$

where $W(q)$ is the work done *by the conservative force* from a chosen reference position to q, and $V(q)$ is potential energy. With consistent reference constants, $W = -V$.

Our task here is different: *given* an action, how does stationarity turn it into an equation of motion?

9 The one new feature: changing position also changes velocity

If

$$q_\varepsilon(t) = q(t) + \varepsilon\eta(t),$$

then differentiating with respect to time gives

$$\dot{q}_\varepsilon(t) = \dot{q}(t) + \varepsilon\dot{\eta}(t).$$

Thus perturbing a trajectory changes *both* its position and its velocity.

That is the only genuinely new complication in the calculation.

Consider

$$S[q + \varepsilon\eta] = \int_{t_0}^{t_1} L(q + \varepsilon\eta, \dot{q} + \varepsilon\dot{\eta}, t)\,\mathrm{d}t.$$

Treat ε as an ordinary number and differentiate. The chain rule gives

$$\boxed{\delta S[q;\eta] = \int_{t_0}^{t_1} \left(\frac{\partial L}{\partial q}\eta + \frac{\partial L}{\partial \dot{q}}\dot{\eta}\right)\mathrm{d}t.}$$

This is the first variation of the action. The first term comes from changing position; the second comes from changing velocity.

10 The crucial step: integration by parts

We would like the entire first variation to be expressed in terms of η, rather than a mixture of η and $\dot{\eta}$.

Why? Because then we can ask what must be true for the result to vanish for every possible shape of η.

Integration by parts gives

$$\int_{t_0}^{t_1} \frac{\partial L}{\partial \dot{q}} \dot{\eta} \, \mathrm{d}t = \left[\frac{\partial L}{\partial \dot{q}} \eta \right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta \, \mathrm{d}t.$$

The boundary term disappears because $\eta(t_0) = \eta(t_1) = 0$.

So our first variation becomes

$$\boxed{\delta S[q;\eta] = \int_{t_0}^{t_1} \left[\frac{\partial L}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) \right] \eta(t) \, \mathrm{d}t.}$$

We have reached the key moment.

The bracket is determined by the proposed trajectory q. But η can be any sufficiently smooth bump that vanishes at the endpoints.

If the bracket were positive somewhere, we could choose a bump supported there and make the integral positive. If it were negative somewhere, we could choose a bump there and make it negative. For the integral to vanish for *every* bump, the bracket must vanish throughout the interval (assuming the usual continuity).

This observation is the **fundamental lemma of the calculus of variations**.

Therefore,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0.}$$

This is the **Euler–Lagrange equation**.

What has happened? We started with one condition on a *whole curve*: its action does not change to first order under any small allowed perturbation. Because the perturbation could be concentrated anywhere, that global condition forced an equation to hold at *each instant*.

11 Return to our familiar particle

For a particle moving in one dimension under a conservative force,

$$L(q, \dot{q}) = \frac{1}{2} m \dot{q}^2 - V(q).$$

The two partial derivatives are

$$\frac{\partial L}{\partial \dot{q}} = m\dot{q}, \qquad \frac{\partial L}{\partial q} = -V'(q).$$

Substituting into the Euler–Lagrange equation,

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\dot{q}) + V'(q) = 0.$$

For constant mass,

$$\boxed{m\ddot{q} = -V'(q) = F(q).}$$

This is Newton's second law. The action formulation and the familiar force formulation describe the same motion for this system.

12 Why this is a satisfying connection

Let us retrace the argument, without calculation.

In ordinary calculus, we perturb a number. If the first-order change in the output vanishes for every small change in that number, we have a stationary point.

In the calculus of variations, we perturb a whole function. If the first-order change in the output vanishes for every small change in that function, we have a stationary function.

The action is a functional because it assigns one number to an entire possible history.

The Euler–Lagrange equation is what we obtain when we express the stationarity of that functional in terms of the position and velocity at each instant.

And the reason the equation of motion is *local* is that we were allowed to perturb the trajectory in an arbitrarily small neighbourhood of any chosen time.

The bridge between the two subjects

$$f'(x_*) = 0 \quad \longleftrightarrow \quad \delta S[q_*; \eta] = 0 \quad \text{for all } \eta$$

The left-hand condition identifies a stationary *number*. The right-hand condition identifies a stationary *curve*. The Euler–Lagrange equation is the differential equation hidden inside the condition on the curve.

A final caution about the intuition

In our earlier discussion, we described stationary action as a balance between the kinetic contribution and the work contribution. That is a helpful motivation, but it is not a claim that each contribution is individually minimized or that they have equal values.

What is balanced is their *first-order response to a change of trajectory*. And because changing position also changes velocity in adjacent moments, the balance becomes an equation involving acceleration.

That is the conceptual achievement of the calculus of variations: it makes the phrase *"no small alteration of the whole path changes the action to first order"* mathematically precise.

End of companion notes